IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-531

1. Topic (12 words max)

SmartTask: Context-Aware Mobile To-Do List for Intelligent Task Management

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	25-26J-531
Year	2025

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The widespread use of digital productivity tools has introduced new challenges, including cognitive overload and notification fatigue. Traditional task management applications typically present a static list of all tasks, regardless of the user's current context, such as location, time, or activity. This can overwhelm users and reduce their ability to prioritize effectively, leading to lower task completion rates [1][2][3]. Additionally, pre-scheduled or manually triggered reminders often result in notification fatigue, where users habitually dismiss irrelevant alerts and risk missing important tasks [4]. Proactively adapting task presentation and providing intelligent reminders that align with a user's real-time context can improve task adherence and reduce the mental effort required for task prioritization [5].

Context-aware computing has matured and been applied successfully in domains such as health monitoring, smart notifications, and location-based services [6][7]. However, a significant gap remains in applying these principles comprehensively to personal task management. Existing solutions primarily focus on a single contextual dimension—for example, location-based reminders for specific locations—or are designed for structured enterprise environments rather than the dynamic and unstructured nature of personal daily life [8]. Furthermore, current applications lack the integration of multiple contextual factors, such as real-time location, device state, activity patterns, and time, to dynamically prioritize and present tasks.

My research aims to fill this gap by developing a user-centric, context-aware to-do list application that intelligently adapts task presentation and notifications based on multiple real-time contextual factors. By doing so, the system seeks to reduce cognitive overload, minimize notification fatigue, and improve task adherence. The goal is to provide a smarter, more intuitive task management solution that aligns with the dynamic nature of modern life, combining effective context detection with adaptive notification strategies to optimize productivity and user well-being.

References

[1] Mark, G., Gudith, D., & Klocke, U. (2008). The cost of interrupted work: More speed and stress. *CHI '08 Proceedings*, 107–110.

[2] Gonzalez, V.M., & Mark, G. (2004). "Constant, constant, multi-tasking craziness": Managing multiple working spheres. *CHI '04 Proceedings*, 113–120.

[3] Bailey, B., & Konstan, J. (2006). Are Informational Notifications Disruptive? *CHI '06 Proceedings*, 331–340.

[4] Magicbell. (2023). Why is Notification Fatigue a Real Problem? Retrieved from https://www.magicbell.com/blog/what-is-notification-fatigue-and-how-to-avoid-it

[5] UXPin. (2019). The Future of UI: Context-Aware Design. Retrieved from https://www.uxpin.com/studio/blog/the-future-of-ui-context-aware-design/

[6] Dey, A.K. (2001). Understanding and using context. *Personal and Ubiquitous Computing*, 5(1), 4–7.

[7] Balaji, S., & Dhas, J. (2013). A survey on context-aware systems. *International Journal of Computer Applications*, 61(18).

[8] The KnowMore Project. (2010). Context-Aware, Proactive Delivery of Task-Specific Information. *12th International Conference on Enterprise Information Systems*.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

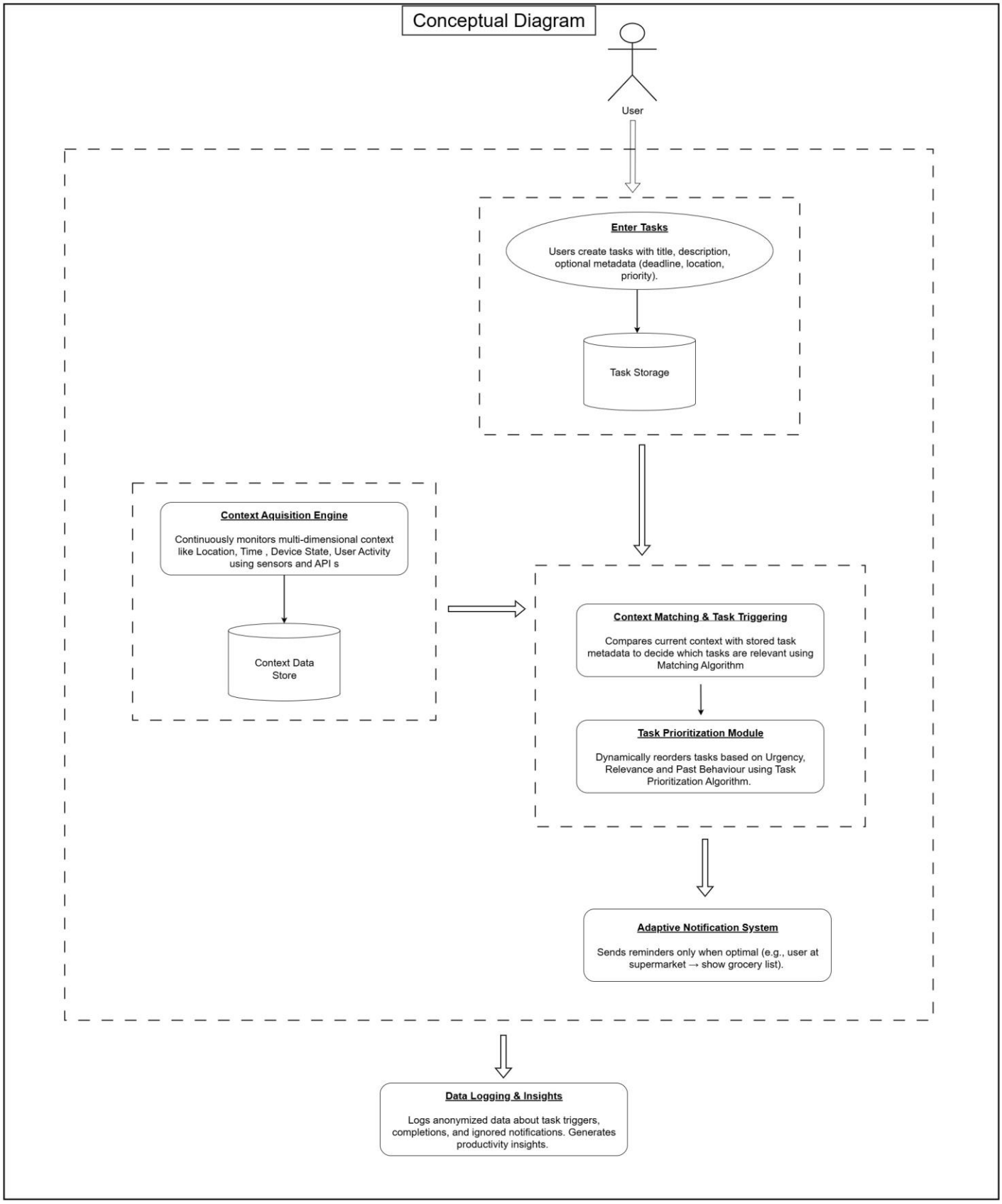
The proposed solution involves developing a **context-aware task management mobile application** that intelligently adapts task presentation and notifications based on the user's real-time context. Unlike traditional to-do list apps that display all tasks statically and rely on fixed reminders, this system leverages multiple contextual inputs to optimize task relevance, minimize cognitive overload, and improve task adherence.

Key Components:

1. **Task Input Module:** Users create tasks with title, description, and optional metadata such as deadline, location, and priority. Metadata is stored in the database for later context matching.
2. **Context Acquisition Engine:** Continuously monitors multi-dimensional context, including location (via Google Maps API geofencing), time, device state (battery, connectivity), and user activity patterns (calendar events, phone motion/activity recognition). Data is stored in the Context Data Store.
3. **Context Matching & Task Triggering:** Compares current context with task metadata to identify relevant tasks, producing a filtered task list for display.
4. **Task Prioritization Module:** Dynamically reorders tasks based on urgency, relevance, and past completion behavior to present the most important tasks first.
5. **Adaptive Notification System:** Sends intelligent reminders and nudges only when the user is in an optimal context for action, reducing notification fatigue.
6. **Data Logging & Insights:** Logs anonymized data on task triggers, completions, and ignored notifications, enabling analysis for productivity insights and improvement of context detection and recommendation algorithms.

High-Level Flow: Task creation → Context acquisition → Context matching → Task prioritization → Adaptive notifications → Data logging for insights and continuous learning.

Area	**How it is done/suggested in previous studies**	**Proposed solution**	**Advantages over the existing one**
Context Acquisition	Limited to single dimensions like time or location; manual input or static reminders	Collects multi-dimensional context: location, time, device state, activity patterns; automatic detection	More comprehensive, reduces user effort, increases task relevance
Task Prioritization	Tasks shown statically; no dynamic adaptation	Dynamically prioritizes tasks based on real-time context and relevance	Reduces cognitive overload and attention residue; improves completion rates
Notifications	Pre-scheduled or manual reminders	Adaptive reminders triggered only under optimal context	Minimizes notification fatigue and increases adherence
User Engagement	No personalization or learning from user behavior	Learns user habits to improve future task recommendations	Creates a personalized experience, improving productivity and usability


Conceptual Diagram
User
Enter Tasks
Users create tasks with title, description, optional metadata (deadline, location, priority).
Task Storage
Context Aquisition Engine
Continuously monitors multi-dimensional context like Location, Time , Device State, User Activity using sensors and API s
Context Data Store
Context Matching & Task Triggering
Compares current context with stored task metadata to decide which tasks are relevant using Matching Algorithm
Task Prioritization Module
Dynamically reorders tasks based on Urgency, Relevance and Past Behaviour using Task Prioritization Algorithm.
Adaptive Notification System
Sends reminders only when optimal (e.g., user at supermarket → show grocery list).
Data Logging & Insights
Logs anonymized data about task triggers, completions, and ignored notifications. Generates productivity insights.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Specialized Domain Expertise

1. **Mobile Application Development:**
   • Proficiency in building cross-platform mobile applications (Android/iOS).
   • Knowledge of mobile SDKs, background services, notifications, and permission management.
   • Experience with optimizing battery usage and app performance for context-aware features.

2. **Context-Aware Computing and Geofencing:**
   • Understanding principles of context-aware systems, including location, time, and activity-based task triggering.
   • Skills in using Google Maps API, Places API, and Geofencing APIs to detect and monitor user location.
   • Knowledge of activity recognition APIs for behavior inference (walking, driving, idle).

3. **Software Development and Data Management:**
   • Proficiency in programming languages and frameworks like Flutter, React Native, Kotlin, Swift, Python, and Java.
   • Experience in integrating secure cloud storage and databases for context and task data.
   • Familiarity with encryption and privacy-preserving techniques for sensitive user data.

Knowledge Requirements

1. **Context Detection and Task Prioritization:**
   • Techniques for combining multiple contextual inputs (location, time, device state, activity) to present relevant tasks.
   • Understanding user behavior modeling to improve relevance and minimize notification fatigue.

2. **Privacy and Security:**
   • Best practices for securing location data and contextual information.
   • Compliance with mobile privacy regulations and data protection standards.

Data Requirements

• Task metadata: location, time, priority.
• User context: GPS/location, device state, motion/activity sensors.
• Historical context and behavior data for pattern recognition and task relevance optimization.

8. Objectives and Novelty

Main Objective :
To develop a context-aware to-do list application that intelligently presents tasks and sends notifications based on the user's real-time context (location, time, activity, and device state) to improve task adherence and reduce cognitive overload.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT21228612 - De Silva S.J.W.	Context-aware task triggering	Implement geofencing, time-based triggers, and device/activity state detection to show relevant tasks	Integrates multi-dimensional context (location, time, activity, device state) rather than relying on a single dimension as in previous studies, ensuring higher task relevance and reduced manual effort
	Adaptive notifications and nudges	Design intelligent notifications that prompt task completion when context is appropriate	Moves beyond pre-scheduled/manual reminders by triggering notifications only under optimal conditions, reducing fatigue and increasing adherence
	Task prioritization and relevance	Dynamically prioritize tasks based on user context, urgency, and historical behavior	Shifts from static task lists to real-time dynamic reordering using contextual data and behavioral patterns, improving focus and completion rates
	Data logging for insights	Collect anonymized context and task completion data for pattern analysis	Unlike existing apps, provides continuous learning by logging and analyzing user patterns, enabling personalization and contributing to long-term productivity research

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	**Description**
IT21228612 - De Silva S.J.W	**Context-Aware Task Triggering:** Designing and implementing logic to detect user context (location, time, device state) and trigger tasks appropriately. **Applies software engineering principles in modular design, context detection algorithms, and integration with mobile sensors.**
	Adaptive Notifications and Nudges: Developing dynamic notification mechanisms that respond to user context and behavior**. Demonstrates knowledge of event-driven programming, user-centered design, and SE best practices for usability and engagement.**
	Task Prioritization and Relevance: Implementing algorithms to rank and present tasks based on contextual relevance and user preferences**. Applies software engineering concepts in algorithm design, data modeling, and system optimization for better user experience.**
	Data Logging for Insights: Capturing and storing contextual, behavioral, and usage data securely for analysis**. Highlights SE expertise in database design, secure data handling, cloud integration, and analytics for continuous improvement.**

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr	Suresh Niroshan	Fernando	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

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Topic Assessment Form

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Recommended.

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	✓
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

* Check how you are going to check the success of the project. (How you are going to demonstrate?)

Staff Member's Name	Signature
Suresh Fernando	2025/9/19.

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.

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